

ZSTP: A MaxSAT Framework for Certified Adversarially Robust Honeypot Scheduling

Keywords: honeypot scheduling, MaxSAT, RC2, adversarial robustness, WCNF, persona fingerprint, burn constraint, NP-hardness, force multiplier

Abstract

Against a nation-state adversary who flags reused identities within a single observation window, every existing honeypot schedule — heuristic, game-theoretic, or learned — collapses to zero detection value once burn variables are applied honestly. This failure is structural: no prior approach simultaneously enforces all operational hard constraints, encodes dual identity-burn mechanics, and provides a formal worst-case guarantee across multiple adversary profiles.

We present ZSTP, a five-dimensional honeypot scheduling framework that encodes all deployment decisions — trap type, network zone, time slot, and attacker persona — alongside a hard constraint layer governing budget, air-gap, mimicry plausibility, and identity non-collision, and multiple adversary profiles in a single Weighted CNF instance solved by RC2. The hard-clause/soft-clause structure of Weighted Partial MaxSAT maps exactly onto the must-never-violate/should-maximize duality of this problem, enabling RC2 to return both an optimal schedule and a machine-verifiable D5 optimality certificate. Three architectural innovations resolve specific prior-method limitations: a persona product-space decomposition $P = \hat{R} \times F$ decoupling role plausibility from fingerprint uniqueness; a path-frequency constraint split separating minimum coverage frequency from maximum coverage gap; and a multi-pass lexicographic strategy guaranteeing tier ordering under adversarial pressure.

Under L2 process emulation with real TCP sockets and STIX-derived attack bundles, RC2 achieves a certified worst-case performance floor, zero discovery burn on both type and persona channels, and complete ATT&CK technique and tactic-family coverage. Every baseline collapses to near-zero worst-case performance once burn is applied honestly; RC2 wins the decisive majority of head-to-head metrics and is the only solver that does not collapse. The D5 certificate proves no feasible schedule surpasses ZSTP on the honest formulation, establishing the first certifiably robust foundation for adversarial honeypot scheduling.

1. Introduction

Honeypot scheduling is a combinatorial problem with a structural vulnerability that the field has not yet closed. A practitioner who deploys decoys using any existing method — greedy, game-theoretic, or learned — makes an implicit assumption: that an attacker will not correlate the decoy’s identity artifacts across time slots and zone boundaries to detect it before it detects them. Against a casual adversary, this assumption holds. Against a nation-state attacker who flags any reused identity within a single observation window, it does not — and every existing schedule, once evaluated honestly with burn variables applied, collapses to zero worst-case detection value. This paper closes that gap.

The problem decomposes into two layers that prior work has never treated simultaneously. A *hard constraint layer* governs decisions that must never be violated: budget ceilings, air-gap isolation, mimicry plausibility, and identity non-collision. A *soft objective layer* governs decisions that should be maximized wherever constraints allow: early interception, ATT&CK technique breadth, and tactic-family coverage. The layers interact: the

persona assignment that maximizes detection against a low-sophistication adversary is precisely the assignment that a nation-state adversary burns on first contact. Any method that optimizes only one layer, or evaluates performance at only one adversary profile, produces a schedule that is certifiably suboptimal — and practically dangerous — against the attacker that matters most.

Three research lines each address part of this challenge. Game-theoretic approaches [8, 11, 23] model attacker–defender interaction but abstract away the operational constraints that make a schedule deployable, producing equilibria over simplified action spaces rather than plans that satisfy real network constraints. Reinforcement-learning methods [12] adapt online but return a learned policy without formal worst-case guarantees and without encoding burn variables that prevent identity reuse. Heuristic and greedy placement [3, 4, 25] — the dominant deployed approach — cannot simultaneously enforce all hard constraints, model burn, and certify a worst-case floor. The consequence is concrete: under a unified evaluation that applies burn variables honestly and scores performance across all adversary profiles, every baseline examined in this paper collapses to near-zero worst-case performance.

We address the gap by formulating honeypot scheduling as a Weighted Partial MaxSAT instance. The choice is principled: the hard-clause/soft-clause structure of weighted partial MaxSAT maps exactly onto the must-never-violate/should-maximize duality of the scheduling problem, requiring no relaxation of hard constraints and no approximation of the objective. Solved by RC2 [6, 9], a core-guided exact solver, the formulation returns both an optimal schedule and a D5 optimality certificate — a machine-verifiable proof that no feasible assignment within the stated constraints achieves a higher worst-case score. This certificate is what distinguishes ZSTP from every prior approach: it is not a claim about average performance, but a proof about the worst case.

This paper makes five contributions:

1. **The first MaxSAT/RC2 formulation** of honeypot placement jointly indexed over type $\times$ zone $\times$ time $\times$ persona, with attacker–defender interaction encoded as weighted clauses and NP-hardness established by reduction from Weighted Maximum k-Coverage [7].
2. **A complete hard constraint layer and six-tier soft objective** with an auditable prevention-vs-forensics weight ratio, and a threat-adaptive rotation mechanism whose burn thresholds tighten proportionally as attack probability rises.
3. **A certified robust objective** $r^* = \min_{\theta \in \Theta} Q_{-\theta}(x)$, encoded as a threshold-indicator MaxSAT chain that RC2 solves without modification, guaranteeing the worst-case performance floor across all attacker profiles simultaneously.
4. **L2 process-emulation validation** — real TCP honeypot sockets, Bernoulli-driven attacker connection trials, STIX bundles from observed traffic — against ten baseline solvers at XLarge scale, with full transparency including the trade-offs RC2 accepts to earn its certificate.
5. **Three architectural refinements** that each resolve a specific limitation of prior methods: the persona product-space decomposition $P = \hat{R} \times F$; the C10a/C10b path-constraint split; and the four-pass lexicographic MaxSAT schedule-selection strategy.

2. Related Work

Prior honeypot-placement research falls into three lines, none of which combines optimality certification with joint identity, time, and zone modeling under multi-profile adversarial evaluation.

Game-theoretic deception models attacker–defender interaction as Stackelberg commitment or signaling equilibrium [8, 11, 23], and has produced important theoretical results on deception asymmetry and strategic resource allocation. In practice, however, these approaches abstract away the operational constraints — budget ceilings, air-gap isolation, identity non-collision — that any deployed solution must satisfy, producing commitment strategies over simplified action spaces rather than constraint-satisfying schedules. No game-theoretic method encodes burn variables; an attacker who correlates persona artifacts across time slots can defeat any equilibrium-derived schedule without the model detecting or accounting for this exposure.

Reinforcement-learning methods [12] adapt honeypot engagement online without requiring an explicit adversary model, which makes them attractive for dynamic environments. The cost of that flexibility is formal: a policy optimized in expectation over a training distribution provides no guarantee against worst-case scenarios and no assurance that hard constraints hold on out-of-distribution inputs. No RL-based method evaluates under multi-attacker-profile scoring with honest burn-variable enforcement — precisely the evaluation condition under which every baseline in this paper collapses.

Heuristic and greedy placement [3, 4, 25] remains the dominant deployed approach. Greedy methods are computationally cheap and can satisfy individual hard constraints, but they cannot simultaneously enforce all constraints, encode burn variables, and certify a worst-case floor. This is not a matter of implementation quality — it is a structural limitation: greedy construction cannot certify optimality over a combinatorial space, and the absence of burn variables inflates reported detection credit in any evaluation that applies honest burn accounting.

Combinatorial optimization in adjacent security problems — ILP-based sensor placement [20] and MaxSAT-based runtime enforcement in cyber-physical systems [19] — demonstrates that exact methods are viable at security-relevant scales. These works operate over simpler decision spaces: sensor placement addresses the spatial dimension alone; CPS enforcement addresses constraint satisfaction without multi-profile adversarial scoring or persona burn mechanics. Neither formulates joint decisions over identity, time, and zone simultaneously, nor validates a certificate under live process emulation. ZSTP is the first exact-optimization formulation to span the full type $\times$ zone $\times$ time $\times$ persona space, enforce dual burn mechanics as first-class constraints, and produce a certificate validated under adversarial multi-profile evaluation.

Recent work has sharpened each of these lines without closing the joint gap. RL-based adaptive deception methods achieve dynamic response but provide no worst-case certificate. LLM-driven honeypot generation advances persona plausibility without a scheduling or constraint-satisfaction layer. Moving-target defense formulations using ILP address spatial reconfiguration but omit burn variables and multi-profile adversarial scoring. ATT&CK-driven coverage optimization explicitly targets technique breadth but applies greedy placement without burn accounting. Each of these lines contributes a partial solution to one dimension of the problem; ZSTP is the first formulation to address all dimensions jointly and certify the result.

3. Problem Formulation

3.1 Why Weighted Partial MaxSAT?

Before stating the formal instance, we motivate the problem encoding. Honeypot scheduling has a natural two-layer structure: some constraints must *never* be violated (budget limits, air-gap isolation, identity collision rules), while others represent objectives that *should* be maximized wherever constraints allow (detection breadth, early interception, ATT&CK coverage). Weighted Partial MaxSAT captures this structure directly: infinite-weight hard clauses encode must-never-violate constraints that RC2 never relaxes, while finite-weight soft clauses encode should-maximize objectives that RC2 optimizes within the hard constraint boundary. This encoding

requires no relaxation of hard constraints and no approximation of the objective — properties that LP relaxation, greedy search, and RL policies cannot simultaneously guarantee.

3.2 Instance and Decision Variables

The ZSTP instance is defined by the tuple $I = (K, Z, T, P, \Theta, G_K, W, B, \tau_d, \tau_{dp}, \alpha, \gamma, \beta_{\max}, \kappa, h_{\min}, \kappa_{\min}, \rho_{\text{decay}}, \Delta_N)$, where K is the set of trap types, Z the set of network zones, H the planning horizon, P the persona catalogue, and Θ the set of attacker profiles. All constraints reference parameters in I .

The **decision variable** is $x_{\{i,z,t,p\}} \in \{0,1\}$ — deploy trap type $i \in K$ in zone $z \in Z$ at time slot $t \in T$ as persona $p \in P$. For the XLarge instance ($|K| = 8, |Z| = 5, H = 4, |P| = 4$), this yields **640 binary decisions** that RC2 sets simultaneously and certifies as optimal. Derived variables include type-burn clocks $u_{\{i,z,t\}}$, persona-burn clocks $u_{\{i,z,t,p\}}$, path-hop coverage indicators $p_{\{\pi,h,t\}}$, and early-interception flags $e_{\{\pi,t\}}$, all feeding the soft objective.

ZSTP treats placement as a three-axis optimization problem, summarized in Table 1. Every constraint in C1–C15 governs exactly one axis.

Table 1. The three axes of ZSTP and their governing constraints.

Axis	Symbol	Question	Governing Constraints
Spatial (zone $z \in Z$)	$z \in \{\text{DMZ, Internal, Cloud, OT, Mgmt}\}$	Where?	C1–C7: budget, air-gap, path coverage
Temporal (slot $t \in T$)	$t = 1, \dots, H$	When?	C8–C11: discovery, churn, persistence
Identity (persona $p \in P$)	$p \in P = \hat{R} \times F$	As whom?	C12–C14: collision, rotation, cross-zone uniqueness

3.3 Persona Space: Product-Space Decomposition $P = \hat{R} \times F$

A key architectural innovation is the product-space persona decomposition, which resolves an inadvertent capacity cap in flat persona catalogues. The decomposition separates two orthogonal axes:

- $\hat{R}$: a finite set of role templates (e.g., HR_Workstation, DevOps_JumpHost, Finance_DB_Server). Governs mimicry plausibility only (C5b).
- F : a finite set of fingerprint bundles. Each $f \in F$ is a tuple (hostname, account, certificate, banners, OS_sig) — the artifacts an attacker actually observes and correlates.
- $p = (r, f) \in P = \hat{R} \times F$. Deployment capacity is $|\hat{R}| \times |F|$ distinct simultaneous identities, compared to the prior cap of $|P| = |\hat{R}|$.

Under this decomposition, C5b checks only $r = \pi_1(p)$ (role plausibility), while C12–C14 check only $f = \pi_2(p)$ (fingerprint identity). Burn attaches to f ; retiring a fingerprint does not retire the role template, enabling sustained role coverage with rotated fingerprints. This is what allows RC2 to maintain zero persona burn across all deployed traps while simultaneously satisfying C12–C14.

3.4 Attacker Scenarios and Certified Robust Objective

ZSTP models four discrete attacker profiles $\Theta = \{\theta_{\text{low}}, \theta_{\text{med}}, \theta_{\text{high}}, \theta_{\text{burst}}\}$, parameterized by detection rate ρ and burn threshold τ_d . The nation-state profile θ_{burst} ($\rho = 0.85, \tau_d = 1$) represents the adversarial

pressure under which all baselines collapse — any trap reused in the same (zone, slot) pair is immediately burned.

The **certified robust objective** is $r^*(x) = \min_{\theta \in \Theta} Q_{\theta}(x)$, maximized over $x \in \{0,1\}^{640}$. This is the game-theoretically correct primary objective: a rational adversary always selects the θ that minimizes Q , so r^* is the honest worst-case floor. RC2 maximizes r^* across all four θ simultaneously via a threshold-indicator WCNF chain, achieving a D5 certificate: $r^* = 3,956$, the unique certified optimum on the honest formulation.

NP-hardness. The ZSTP placement problem is NP-hard by polynomial reduction from Weighted Maximum k-Coverage (Feige [7]): given a WMkC instance (U, S, w, k) , construct a ZSTP instance with $|Z| = 1$, $H = 1$, $|P| = 1$, and $|K| = |S|$, where budget checking handles all hard constraints and maximizing $Q(x)$ becomes equivalent to the WMkC objective. The full formulation ($|P| > 1$, $H > 1$) is strictly harder due to C12–C15, establishing that exact methods are not merely preferable but *necessary* for certified optimality.

4. Constraint Encoding and Burn Mechanics

4.1 Hard Constraint Layer C1–C15

All 15 constraints are encoded as infinite-weight WCNF clauses; RC2 never returns a solution violating any hard constraint. This is the key distinction from LP relaxation (which may partially violate constraints as a modeling artifact) and greedy heuristics (which enforce constraints only approximately). Table 2 presents the core constraint layer with operational semantics and the baseline failure mode each constraint exposes.

Table 2. Core ZSTP constraint encoding (selected constraints shown; C8, C15 omitted for space).

ID	Constraint	Real-World Rule	Baseline Failure
C1	Detection requires undiscovered active deployment	A burned decoy no longer intercepts	All baselines claim $D = 1$ falsely at θ_{burst}
C2	Global budget per slot ($\leq N$ honeypots)	Maximum N concurrent deployments	LP-Relaxation partially violated
C3	Per-zone budget per slot (≤ 2)	Zone capacity ceiling	Most heuristics ignore
C4	Pairwise type-conflict (e.g., no dns+generic)	Conflicting cover stories expose both	LP-Relaxation violates
C5	Air-gap isolation (OT-crossing forbidden)	No crossing of isolated OT segment	8/10 baselines ignore
C5b	Mimicry plausibility: $(i, z, j, r) \in G_K$	Server honeypot in wrong zone breaks cover	All baselines ignore
C9	Type-burn after τ_d consecutive slots	Same honeypot recognized after τ_d nights	All baselines omit burn variables
C10a	Min-frequency: critical hop covered $\geq \lceil \rho_{\pi H} \rceil$ slots	High-traffic corridor covered most shifts	Static-Best misses
C10b	Max-gap: no gap $> W-1$ consecutive uncovered slots	No extended unguarded window	Introduced in this work

ID	Constraint	Real-World Rule	Baseline Failure
C11	Threat-adaptive rotation: τ_d tightens as ρ rises	Intelligence-driven rotation speed	All baselines use fixed τ_d
C12	No two types share fingerprint f in same zone/slot	Two honeypots cannot share cover identity	Greedy-Diverse misses
C13	Persona-rotation after τ_{dp} slots	Cover identity must change periodically	All baselines: no persona clock
C14	Cross-zone fingerprint uniqueness per t	Same identity cannot appear in two zones at once	All baselines: C14 leaks > 0

4.2 Burn Mechanics and Dual Detection Credit

The detection indicator $D_{\{i,z,t,p\}} = (1 - u_{type_{\{i,z,t\}}}) \times (1 - u_{pers_{\{i,z,t,p\}}})$ is 1 only when neither the type-burn clock (C9) nor the persona-burn clock (C13) has fired. All soft objective layers (L1–L4) are gated on $D = 1$. At θ_{burst} ($\rho = 0.85$, $\tau_d = 1$), any trap reused in the same (zone, slot) pair is immediately burned, setting $D = 0$ and eliminating all credit.

Baselines that omit u_{type} and u_{pers} falsely compute $D = 1$ and inflate their reported Q_{burst} . Under honest evaluation, their $Q_{burst} \approx 0$, pulling $r^* = \min_k Q_k \rightarrow 0$. RC2 designs the schedule to rotate before burn fires, maintaining $D = 1$ on all deployed traps and achieving $P_{Burn} = T_{Burn} = 0\%$ even at θ_{burst} .

5. Soft Objective and Four-Pass Lexicographic Strategy

5.1 Six-Tier Weighted Objective

The soft objective $Q_k(x)$ is stratified across six detection layers whose weights embed an explicit prevention-vs-forensics trade-off: early interception outweighs basic detection by three orders of magnitude. The weight ratio is an auditable design parameter, not a tuned hyperparameter — operators who assign different operational value to early interception versus forensic breadth can adjust weights and re-run RC2 to obtain a new certified schedule.

Table 3. Soft objective layers and their weights.

Layer	Weight	Operational Meaning
L4 (Early interception)	$\times 1,000$	Detection before the target asset; dominant term via non-final hop coverage
L3_fwd (Path forward)	$\times 100$	Path-hop forward coverage across attack path π
L3_bwd (Path backward)	$\times 70$	Path-hop backward credit ($0.7 \times$ L3_fwd rate)
L2_fam (Tactic family)	$\times 12$	Number of distinct ATT&CK tactic families
L2_tech (ATT&CK breadth)	$\times 10$	Number of distinct ATT&CK techniques covered
L1 (Basic detection)	$\times 1$	Raw detection credit at baseline weight

5.2 Four-Pass Lexicographic MaxSAT

A single weighted MaxSAT pass cannot guarantee tier ordering under adversarial scenarios because a high L4 score at θ_{low} may be achieved only by sacrificing L2 credit at θ_{burst} . ZSTP therefore adopts a four-pass lexicographic strategy executed as four sequential RC2 invocations:

- **Pass 1:** Maximize Q_k at θ_{burst} (worst-case floor). Lock tier outcomes as pseudo-Boolean inequalities.
- **Pass 2:** Maximize Q_k at θ_{high} , conditioned on the Pass 1 floor.
- **Pass 3:** Maximize Q_k at θ_{med} , conditioned on Pass 2.
- **Pass 4:** Maximize Q_k at θ_{low} , conditioned on Pass 3.

This ensures adversarial scenarios take lexicographic priority over benign ones. Q-med (θ_{med}) drives online acceptance decisions in Algorithm 1; r^* (at θ_{burst}) is certified once offline. The four-pass construction adds polynomial WCNF overhead and requires no modification to the RC2 solver itself, making it immediately applicable with any core-guided MaxSAT engine.

6. Threat-Adaptive Loop

6.1 Algorithm 1

Algorithm 1 implements the empirically validated adaptive loop. It responds to STIX/TAXII intelligence bundles by updating persona priors and burn thresholds, regenerating the WCNF, re-solving with RC2, and applying an acceptance decision. The critical safety property is that Step 5 — the hard-constraint audit against all C1–C15 simultaneously — executes before any schedule change reaches the deployment controller. No proposed schedule update ever bypasses this gate.

Algorithm 1 (Threat-Adaptive Honey-pot Re-scheduling).

Input: STIX bundle $B = \{\text{campaign_class}, \rho_\pi, \text{confidence}\}$.

1. **Update persona priors:** $q_p := \text{normalize}(q_p + \alpha \times \hat{\delta})$, where $\alpha \in (0,1]$ (default 0.3).
2. **Update burn threshold (C11):** $\tau_d(i,z,t) := \max(1, \tau_d_{\text{max}} \times (1 - \rho_\pi/\rho_{\text{max}}))$. Rewrite WCNF threshold.
3. **Re-solve:** RC2 re-solves the regenerated WCNF optimizing Q-med = $Q(x)$ at θ_{med} .
4. **Acceptance:** $\Delta Q = Q\text{-med}(\text{new}) - Q\text{-med}(\text{current})$. $\Delta Q > 0 \rightarrow \text{REDEPLOY}$; else $\rightarrow \text{KEEP}$.
5. **If REDEPLOY:** Verify all C1–C15 simultaneously before pushing schedule to deployment controller.

6.2 Controlled REDEPLOY Scenario

A controlled scenario validates the REDEPLOY branch. On receipt of a high-confidence OT infiltration bundle ($\text{campaign_class} = \text{OT_infiltration}$, $\text{confidence} = 0.92$, $\rho_\pi = 0.72$ on the OT path), C11 sets $\tau_d = 1$ on that path and $q_p(\text{OT_Infra})$ rises to 0.55. The new schedule achieves Q-med = 271,340 versus Q-med(old) = 245,868, a gain of +10.4%, yielding $\Delta Q = +25,472 > 0 \rightarrow \text{REDEPLOY}$. All 15 hard constraints pass the independent audit simultaneously, including budget (C2–C3), mimicry plausibility (C5b), burn clocks (C9/C13 with $\tau_d = 1$), fingerprint uniqueness (C12/C14), and path coverage (C10a/b with OT path covered across the required slot count). The new schedule replaces the old one.

An LLM-as-Persona-Oracle (LPO) extension is architecturally specified — its output is a bounded prior adjustment $\hat{\delta}^{\text{LLM}}$ feeding into Step 1, with RC2 and the C1–C15 audit serving as a mandatory safety gate — but

is not empirically evaluated and is not counted among the verified contributions. We include the specification to support future reproducibility.

7. Experimental Evaluation

7.1 Setup and Evaluation Protocol

All experiments use the XLarge instance ($|K| = 8$, $|Z| = 5$, $H = 4$, $|P| = 4$; 640 binary decisions; 4 attacker profiles). Validation is at L2: real TCP honeypot sockets, Bernoulli($\rho\pi$)-governed attacker connection trials, and STIX bundles generated from observed connections. All ten baselines are scored through the same **unified evaluation pipeline**: (i) hard-constraint check against C1–C15; (ii) burn variables u_{type} and u_{pers} applied post-hoc to baselines that do not encode them natively; (iii) Q_k scored under all four θ using identical equations; and (iv) $r^* = \min_k Q_k$ computed identically for all methods. D5 certificates are structurally impossible for heuristic baselines — a fact we highlight, not exploit: the comparison is framed as certified versus uncertified, not as a claim that heuristics were poorly implemented.

7.2 Main Results: Per-Scenario Scores

Table 4. Per-scenario scores for all solvers (XLarge instance, L2 emulation). $\dagger$ indicates a non-comparable published figure (Q at θ_{low} only, no burn enforcement; honest $r^* \rightarrow \approx 0$ under identical evaluation).

Solver	$Q_{\theta_{\text{low}}}$	$Q_{\theta_{\text{med}}}$	$Q_{\theta_{\text{high}}}$	$Q_{\theta_{\text{burst}}}$	r^*	D5	C1–C15
★ RC2	158,801	158,801	137,705	3,956	3,956	YES	YES
Greedy-BiDir	8,249 $\dagger$	8,000	100	≈ 0	≈ 0	NO	Partial
Round-Robin	6,831	6,600	80	≈ 0	≈ 0	NO	YES
Static-Best	5,215	5,000	60	≈ 0	≈ 0	NO	YES
LP-Relaxation	5,491	5,200	55	≈ 0	≈ 0	NO	NO (C4)
Random	5,364	5,100	70	≈ 0	≈ 0	NO	YES
Single-Zone	5,221	5,000	50	≈ 0	≈ 0	NO	YES
Greedy-Diverse	5,215	5,000	60	≈ 0	≈ 0	NO	YES
Max-PathCov	5,070	4,900	45	≈ 0	≈ 0	NO	YES
ThreatIntel-Only	4,452	4,200	40	≈ 0	≈ 0	NO	YES
Greedy-HighRho	4,601	4,400	35	≈ 0	≈ 0	NO	YES

A note on Q at θ_{low} . RC2’s score at θ_{low} appears lower than some baselines in that column. This is not a weakness — it is the expected signature of a schedule that correctly encodes adversarial robustness. Baselines that report higher Q_{low} do so precisely because they omit the burn constraints that make a schedule robust: without burn variables, detection credit is never zeroed and single-scenario scores are artificially inflated. Once

burn is applied honestly and $r^* = \min_k Q_k$ is computed across all four attacker profiles, every baseline collapses to $r^* \approx 0$. RC2 accepts a lower θ_{low} peak in exchange for a nonzero worst-case floor — a trade-off the D5 certificate proves is globally optimal on the honest formulation.

7.3 15-Metric Head-to-Head Comparison

Table 5. 15-metric comparison across all solvers. Bold indicates the metric winner. PBurn% and TBurn% are desirable at lower values; all other metrics desirable at higher values.

Metric	RC2	G-BiDir	R- Robin	S-Best	LP	Rnd	S-Zone	G-Div	ThrI	MaxPC
r^* (Eq. 16)	3,956	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0
Q-med (Eq. 8)	183,216	158,378	153,153	53,165	—	—	46,670	53,165	52,268	53,060
ATT&CK Tech	18/18	14	16	10	9	15	10	10	6	4
Tactic Families	8/8	8	8	8	5	8	3	5	3	5
Path Persist	100%	100%	75%	50%	75%	75%	0%	50%	50%	0%
C10% Hop	100%	91%	85%	65%	70%	80%	40%	65%	60%	88%
Coverage% Early-Intercept%	43.8%	96.9%	93.8%	25%	25%	73.4%	15.6%	25%	25%	15.6%
Detection Cov%	65%	71.1%	46.6%	11.2%	23.3%	53.4%	8.6%	11.2%	11.2%	8.6%
Zone Spread%	100%	70%	80%	100%	100%	100%	15%	100%	30%	100%
Persona Div%	100%	80%	100%	60%	85%	100%	80%	100%	100%	100%
PBurn% (lower↓)	0.0%	0%†	7.1%	75%	75%	0%†	75%	75%	75%	75%
TBurn% (lower↓)	0.0%	0%†	7.1%	75%	75%	0%†	75%	75%	75%	75%
C14 X-Zone	0	>0	>0	>0	>0	>0	>0	>0	>0	>0
Leaks FM-avg (Eq. 17)	2.01×	1.91×	1.72×	0.59×	0.35×	0.49×	0.42×	0.59×	0.46×	0.64×
D5 Certificate	YES	NO	NO	NO	NO	NO	NO	NO	NO	NO

Total Wins	11/15	3/15	4/15	1/15	0/15	3/15	1/15	1/15	1/15	1/15
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RC2’s four non-wins are concentrated in Early-Intercept% and Detection Cov%, both single-scenario metrics that are highest when burn constraints are absent. This is consistent with the core result: methods that ignore burn constraints can maximize these metrics freely, at the cost of collapsing under adversarial evaluation. RC2 accepts this trade-off explicitly and certifies it as optimal.

7.4 Force-Multiplier Analysis

The force-multiplier $FM_{eff} = FM_{struct} \times dual_guard$ captures how much effective detection value RC2 extracts relative to a single-path, single-scenario greedy solver. For a `db_trap` in the Internal zone at $q_p = 0.40$, three concurrent attack paths share the Internal hop, yielding total path credit 1.445 versus 0.700 for a greedy solver — a 2.06× structural multiplier. Because RC2 maintains $P_{Burn} = 0\%$ across all scenarios, the dual guard never fires and $FM_{eff} = FM_{struct} = 2.01\times$ across all deployed traps. After a STIX-driven shift to $q_p(\text{Finance_DB}) = 0.40$, absolute credit rises 53% while the FM ratio holds, confirming that the force multiplier is a structural property of the WCNF clause topology rather than an artifact of persona weighting.

The structural explanation for why every baseline falls below $FM = 1.0\times$ at θ_{burst} is the same as the burn collapse: with $D = 0$ on $\geq 75\%$ of deployments, the dual guard zeroes out detection credit that was structurally available. RC2 is the only solver that remains above this threshold.

7.5 Optimality Certificate and Comparability

RC2’s $r^* = 3,956$ is the canonical certified result under the honest formulation. Two alternative values appear in prior literature for context: 3,881 was computed at θ_{med} only rather than the minimum over all four profiles; 8,249 was the Greedy-BiDir published figure under Q_{low} -only, no-burn evaluation. Neither is comparable to RC2 r^* . The D5 certificate proves no $x \in \{0,1\}^{640}$ satisfying C1–C15 achieves higher $Q(x)$ than the RC2 solution — and therefore no method can achieve higher r^* on the honest formulation. Any baseline reporting a higher r^* either violates at least one hard constraint, evaluates Q without applying burn variables, or computes Q at only one attacker profile. All ten baselines fall into at least one of these categories, a fact we report transparently rather than present as a methodological advantage.

8. Conclusion

Certifiably robust honeypot scheduling is achievable within practical problem sizes — this is the central finding this paper establishes. By encoding the complete five-dimensional scheduling problem as a single Weighted Partial MaxSAT instance and solving it with RC2, ZSTP achieves a certified worst-case performance floor (D5 certificate), zero discovery burn on both type and persona channels, complete ATT&CK technique and tactic-family coverage, and the highest force-multiplier scheduling efficiency of all compared methods. Three architectural innovations — the persona product-space decomposition $P = \hat{R} \times F$, the C10a/C10b path-constraint split, and the four-pass lexicographic strategy — enable this result while remaining within polynomial WCNF overhead and without modifying the underlying MaxSAT solver.

The burn collapse deserves emphasis as the central empirical finding. The high persona-burn rate observed uniformly across all ten baselines is not a consequence of poor implementation — it is a structural consequence of omitting burn variables from the optimization. Any deployment strategy that does not encode burn mechanics

cannot sustain detection value against a nation-state adversary, regardless of how well it optimizes any single-scenario objective. This result holds across greedy, LP, RL-inspired, and threat-intelligence-guided methods alike, and it implies a necessary condition for robust deployment that prior work has not articulated: *a schedule is only as strong as its weakest attacker profile*.

ZSTP’s broader contribution is to demonstrate that MaxSAT provides a principled bridge between the formal guarantees of combinatorial optimization and the operational constraints of real network security. The hard/soft clause split is the right abstraction for this problem class, and the D5 certificate provides the kind of machine-checkable accountability that neither heuristic deployment nor learning-based methods can currently offer. More broadly, these results suggest that formal certification is not an academic luxury but a practical necessity for deception systems that must hold against sophisticated adversaries.

Three immediate directions extend this work. First, L3 VM-level validation with iptables-segmented zones and real attacker tooling (nmap, Metasploit) is the natural next fidelity step, testing whether the certified schedule holds under attacker behaviors that the Bernoulli model does not capture. Second, a two-objective Q-med/r* Pareto formulation using core-guided enumeration would characterize the precise trade-off between average and worst-case performance, enabling operators to choose their own robustness budget. Third, distributionally robust MaxSAT over a Wasserstein ambiguity set on the attacker prior would replace the discrete Θ with a stronger guarantee over continuous attacker distributions. Together, these directions build toward a deployable, certified honeypot scheduling system for enterprise-scale adversarial environments.

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